The University of Azad Jammu and Kashmir, Muzaffarabad

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Computer Networks

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Lab 04

LAN Network Configuration using Switch

Department Of Software Engineering

Lab Title: -

Design and Implementation of a LAN using Switch in Cisco Packet Tracer

1. Objective

To design and implement a Local Area Network (LAN) using 5 PCs and a switch, and to test connectivity between devices using ping command.

2. Tools/Requirements:

- Cisco Packet Tracer
- 5 PCs (End Devices)
- 1 Switch (2960)
- Copper Straight-Through Cables

3. Theory:

A Local Area Network (LAN) is a network that connects computers within a limited area such as a lab, school, or office. A switch is a networking device that connects multiple devices in a LAN and forwards data to the correct destination using MAC addresses.

In this lab, each PC is assigned an IP address from the same network so that they can communicate with each other. The ping command is used to test connectivity between devices.

4. Procedure:

Step 1: Place Devices

- Add 5 PCs and 1 switch on the workspace.

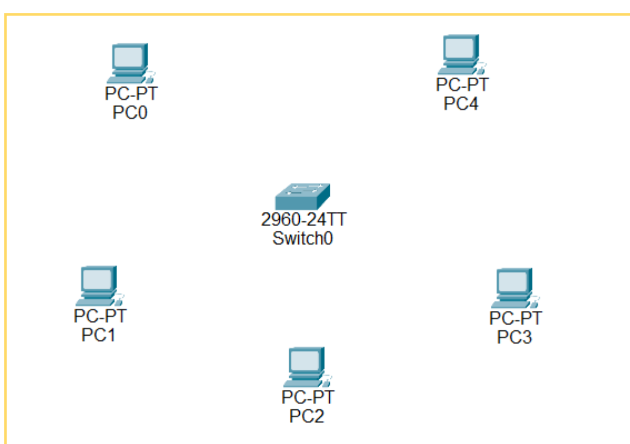


Figure 1.1 Add PCs and Switch

Step 2: Connect Devices

- Use Copper Straight-Through cables to connect each PC to the switch:
    - PC1 → Fa0/1
    - PC2 → Fa0/2
    - PC3 → Fa0/3
    - PC4 → Fa0/4
    - PC5 → Fa0/5

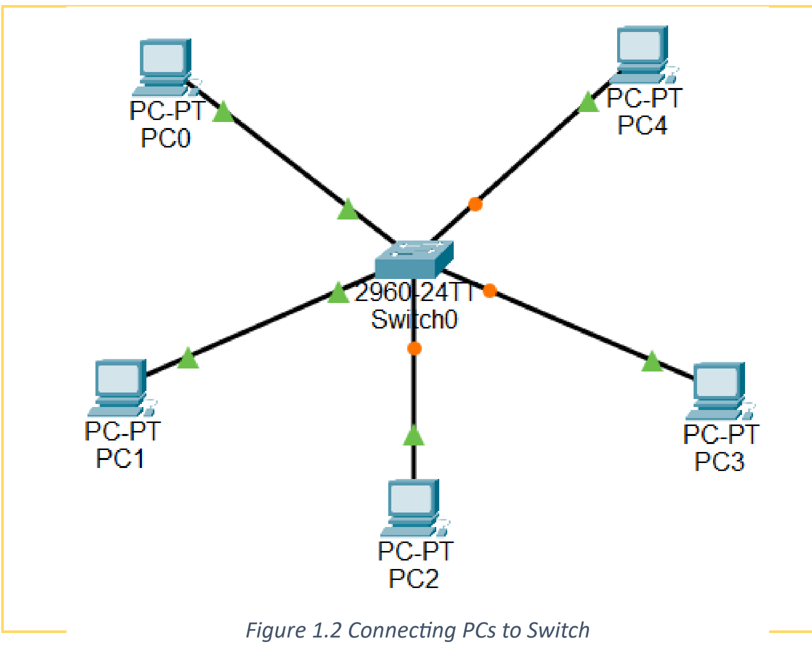
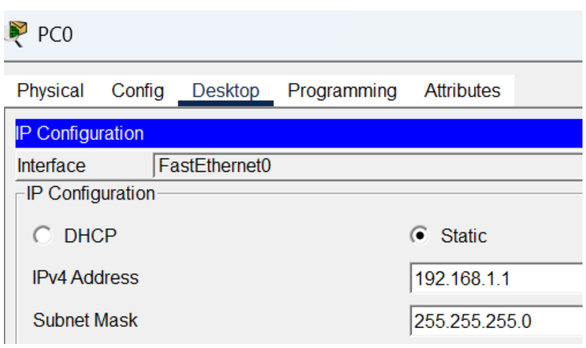


Figure 1.2 Connecting PCs to Switch

Step 3: Configure IP Addresses

Assign the following IP addresses:

- PC0 192.168.1.1
- PC1 192.168.1.2
- PC2 192.168.1.3
- PC3 192.168.1.4



Figurer 1. 3 Assign IP Address

- PC4 192.168.1.5

Step 4: Test Connectivity

- Open Command Prompt on PC1
- Run the command:

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 1.4 Connectivity test

Result:

All PCs successfully communicated with each other. The ping test showed replies with 0% packet loss, confirming that the LAN network is working correctly.

5. Process Cycle:

PC0 send request to Switch the Switch sends to PC4

Then the PC0 send response to Switch and switch sends it

To PC0. In this way one Cycle completes.

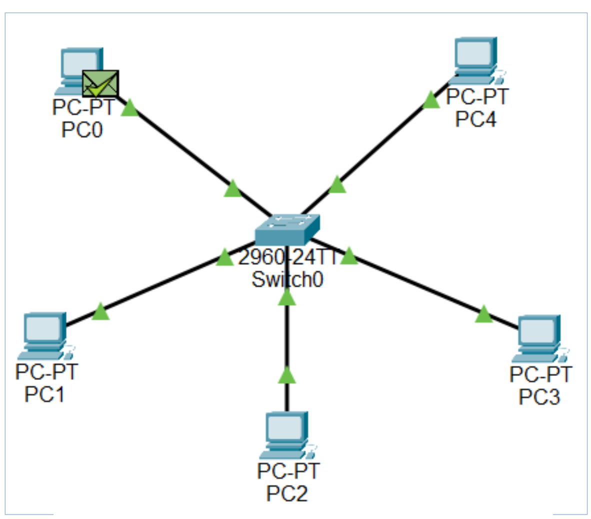
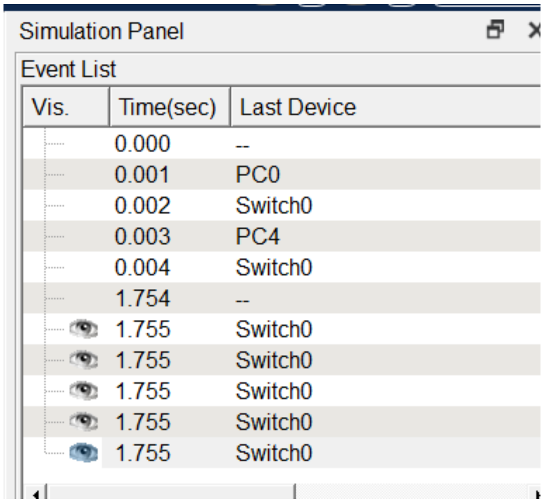


Figure 1.5 Process Cycle

6. Simulation



Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	PC4
	0.004	Switch0
	1.754	--
	1.755	Switch0
	1.755	Switch0
	1.755	Switch0
	1.755	Switch0
	1.755	Switch0

7. Conclusion:

The LAN network was successfully designed and implemented using a switch in Cisco Packet Tracer. All devices were able to communicate with each other, proving that correct IP addressing and proper connections are essential for network communication.